PeerEval: Learning Whose AI Judgments to Trust

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Arena Academic Partnerships Program, Fall 2026 12 months $50,000

Research objective

AI models increasingly judge other AI models: they compare two answers and choose which is better. These verdicts make evaluation scalable, but consensus can reflect shared taste rather than users' preferences. In Arena's study, three judges from one provider rated their own model family 37 percentage points more favorably than human voters did [3]. We propose **human-guided peer review**: a system that learns whose judgments deserve influence from peer evidence and limited human feedback, then identifies when collective judgment needs human review. The scientific question is how to distinguish useful collective expertise from mutually reinforcing bias.

Foundation and preliminary evidence

Our PRD method makes every model both an answer writer and a reviewer: it ranks answers using peer judgments, then updates each judge's influence according to the quality of its own answers [1]. Our PeerEval platform implements a working leaderboard and evaluation tool [2]. A separate pilot covers 1,179 human-labeled battles with four judges. On 794 crowd battles, agreement-based weighting matched 67.8% of human votes, versus 65.7% for equal weighting. These exploratory results motivate the proposed advance: **learn how answer quality, peer agreement, and human feedback jointly determine which judgments to trust**.

Research approach

1. Infer judge reliability from complementary peer signals. We will assemble six to eight judges from different model families. Each will compare anonymous answer pairs in both display orders, reporting an A win, B win, tie, or both answers being bad. Swapped judgments will be aligned to answer identity and averaged, preserving disagreement about the pair. On separate development prompts, we will construct two candidate signals: *answer competence*, measured by how peers rate a model's own answers, and *judgment agreement*, measured by consistency with judges from other families. We will compare both signals with equal weighting, exclude self-ratings, and smooth candidate weights so a weak answerer can still earn influence as a judge. Their ability to predict human preferences will be tested independently of the data used to construct them.

2. Learn how much human evidence should change peer influence. Using a total budget of 50 to 500 human-labeled comparisons for fitting and threshold selection, we will learn an interpretable combination of judge votes: each judge provides smoothed probabilities for the four outcomes, and nonnegative weights that sum to one determine its influence. Fitting initially favors a candidate peer weighting; calibration data determine which signal to retain and how strongly to retain it, including giving it no influence. Thus human evidence can retain useful peer expertise or override a misleading consensus. We will also penalize errors that judges repeatedly make together against human votes, reducing the influence of redundant opinions. For example, three related judges that favor elaborate answers rejected by people should contribute less than three independent endorsements. Comparisons with calibration from equal starting weights will isolate the value of peer evidence. Label-budget curves will show when peer signals reduce the human feedback needed for reliable evaluation.

3. Identify when collective judgment requires human review. A calibrated panel will also provide an interpretable reliability check. We will measure disagreement across model families, sensitivity to leaving out one family, and how different a comparison is from the calibration prompts. Using separate validation data, we will set thresholds that flag cases where confident-looking agreement has repeatedly failed to match human preferences. This directly addresses the failure that agreement alone cannot reveal: several judges can endorse the same answer for the same unhelpful reason. We will test the flagging rule on unseen prompts, later time windows, and newly introduced answer-model families, measuring errors among accepted decisions and the human review required to resolve flagged cases.

On held-out data, we will report how often the system makes an automatic decision and how often those accepted decisions disagree with people. Agreement that predicts low error will be retained; agreement that accompanies high error will trigger review. This converts a leaderboard from a single ordering into an auditable evaluation with a measured human workload. Separate tests of self-exclusion, family balancing, and the shared-error penalty will establish which corrections matter. As a follow-on extension, flagged comparisons that could change an uncertain model ordering will receive additional model or human judgments. This extension will retain random exploration, record selection probabilities, and correct for selective sampling. We will target 25% fewer model calls than a complete panel at matched ranking agreement, counting all acquired human votes and prior-construction costs.

Evaluation, outputs, and feasibility

We will target up to 10,000 Arena battles using public and, subject to agreement, recent data. Baselines cover single judges, equal weighting, original PRD, agreement weighting, and calibration without peer evidence. Outcomes are preference accuracy and log loss, agreement with human-derived model ranks (Kendall correlation), and cost, with paired bootstrap intervals that refit weights. Length-matched accuracy and an equal average over cases where people prefer longer versus shorter answers will diagnose dependence on length, alongside a longer-answer heuristic; these do not identify a causal effect. We will measure headroom over the strongest judge and use repeated votes across display orders to assess human disagreement. Shared blind spots and differing user preferences may limit gains.

Months 1–3 will establish data and splits; months 4–8 will develop weighting and reliability checks; months 9–12 will complete transfer tests, a paper, an open-source PeerEval extension, and reproducible audits. Adaptive acquisition will follow the core results.

Requested support and value to Arena. The $50,000 gift allocates $36,000 to graduate research, including fringe and tuition; $10,000 to APIs and computing; and $4,000 to human annotations. At maximum scale, 10,000 pairs $\times$ eight judges $\times$ two orders require 160,000 judging calls; pilot costs, including prior construction, will determine the affordable scale. Results will identify peer signals that complement human voting and AutoEval [4] and cases warranting human review.

References

[1] Li R, Patel T, Du X. PRD: Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR, 2024. Paper.

[2] PeerEval. https://peereval-ai.github.io/; pilot results. 2026.

[3] Arena Team. Self-Preference of AI Judges. September 30, 2026. Article.

[4] Arena Team. Introducing AutoEval to the Arena leaderboards. July 30, 2026. Article.